QUESTION PAPER

Geometry Practice Test (KFEL)

GRADE 10

EASY

Algebra | Calculus | Geometry

MAXIMUM MARKS	QUESTIONS	TIME ALLOWED	SECTIONS
33	7	14 Minutes	4

Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS

1. Read all questions carefully before answering.
2. All questions are compulsory.
3. Ensure your handwriting is neat and legible.

SECTION A — Algebra - Multiple Choice

1 Question | 5 Marks

Choose the correct answer from the given options.

1 Which of the following is a correct statement about solving a system of linear equations? — 5 Marks

(A) It can only be solved by substitution.

(B) It can be solved using substitution, elimination, or matrix methods.

(C) It always has exactly one solution.

(D) It can only be solved graphically.

SECTION B — Calculus - Multiple Choice

2 Questions | 20 Marks

Choose the correct answer from the given options.

2 What is the derivative of the function f(x) = sin(x)? — 10 Marks

(A) -sin(x)

(B) cos(x)

(C) -cos(x)

(D) sin(x)

3 Which of the following statements correctly describes the relationship between a function and its derivative at a specific point? **10 Marks**

(A) The derivative gives the average rate of change over an interval.

(B) The derivative at a point indicates the slope of the tangent line to the graph at that point.

(C) The derivative is always positive for increasing functions.

(D) The derivative is the same as the original function when the function is a straight line.

SECTION C — Geometry - Problem Solving

3 Questions | 6 Marks

Solve the problem showing all steps.

4 A right triangle has a base length of 6 cm and a height of 8 cm. Calculate the length of the hypotenuse using the Pythagorean theorem. Show all your steps and find the final length of the hypotenuse in centimeters. **2 Marks**

Working space and answer

5 A right triangle has one leg measuring 9 cm and the hypotenuse measuring 15 cm. Using the Pythagorean theorem, calculate the length of the other leg. Show all your steps and provide the final length in centimeters. **2 Marks**

Working space and answer

6 A right triangle has one leg measuring 5 cm and the hypotenuse measuring 13 cm. Calculate the length of the other leg using the Pythagorean theorem. Show all your steps and provide the final length in centimeters.

2 Marks

Working space and answer

SECTION D — Algebra - Short Answer

1 Question | 2 Marks

Answer in 2-3 sentences.

7 Write the general form of a linear equation in one variable and explain what the constants represent.

2 Marks

Write your answer here

- End of Question Paper -